The structure of groups in free flight

Sizes, cohesion, leadership and partners of the groups labelled by the method of the note *Group flights in free flight*

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1 What we find

What do the groups look like on the archive, and what holds them together? The groups are those labelled by the method of the note *Group flights in free flight*, with its chosen values, $\delta = 2$ min and $q = 0.1$: 21.7% of the paraglider flights and 6.9% of the hang-glider flights fly in a group at some point, in 17 169 episodes (Table 6 of the note). Chance is the null model of Section 5.1 of the note; bands are 5–95% of 200 draws of the dependency blocks, or of the dates where the captions say so. Graph terms are defined in Appendix A, the two models in Appendix B. The report *Group and solo flights* (`analyses/groups/vs-solo`) compares the motion of pilots in a group and alone; some results here refer to it.

1.1 What the groups are

How large are the groups, how long do they last, and how do they change? We count episodes and groups (Section 4 of the note) by number and by duration; an episode lasts from the start of its first group to the end of its last, and its participants are the distinct flights it holds.

Sizes and durations. With the chosen values the median episode lasts 58.2 min and three in four are pairs (Table 1). Episodes last at least τ^*, since a contact must, and larger episodes last longer, so they weigh more in time than in number. Groups have no minimum duration: every arrival or departure of a member starts a new one, so 10% last less than 2.1 min, and

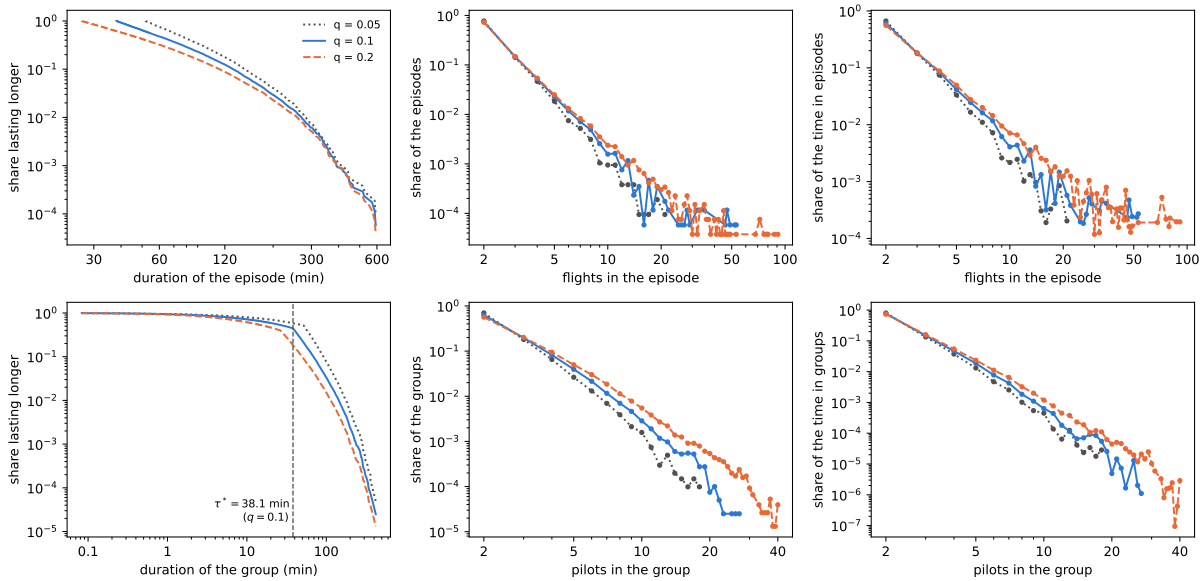


Figure 1: Episodes (top) and groups (bottom), $\delta = 2$ min, whole archive, for three values of q. Left: share lasting longer than a duration. Middle: share of each size. Right: share of the total duration in each size.

	q	number	median duration	2 participants items	2 participants time	5 or more items	5 or more time	largest
episodes	0.05	10 553	74.6 min	77%	67%	3.9%	8.1%	21
	0.1	17 169	58.2 min	74%	61%	5.7%	12.6%	53
	0.2	26 504	44.5 min	73%	56%	7.0%	17.2%	91
groups	0.05	20 275	52.7 min	70%	81%	5.6%	2.3%	18
	0.1	40 051	30.4 min	63%	77%	9.4%	3.5%	27
	0.2	75 953	17.1 min	56%	74%	14.1%	5.0%	40

Table 1: Episodes and groups for three values of q ($\delta = 2$ min). Items: share of the episodes or groups of that size; time: share of their total duration.

the median is 30.4 min. Large groups change quickly, so in time the pairs weigh more than in number. 0.9% of the groups hold a hang glider. Inside a group the edges among the members can change: the 40 051 groups of the chosen values hold 52 856 configurations. A larger q, hence a shorter τ^*, gives more, shorter and larger groups: the shares of the small sizes change little, and the tail moves.

How groups change. A group ends in one of five ways: a *join* (members added), a *leave* (members removed in flight), a *landing* (a removed member lands within 5 s, the grid step), a *split* or a *merge* (Appendix A); a group of two pilots that loses one dissolves. Table 2 gives, by size, the groups ending per hour spent at that size.

- **Most episodes never change:** 74% of the 17 169 episodes are one group; 5.0% hold a split, 4.4% a merge.
- **Episodes last; groups change.** 12% of the episodes last 2 h or more. A group lasts 41 min with 2 pilots and 4 min with 8 or more: joins, leaves, splits and merges all grow with size, so a large group changes members every few minutes.

- **Groups break in flight:** 91% of the 18 376 groups that dissolve do so in flight, 9% at a landing; a pair ends mostly because the two separate. 34% of the 18 218 groups with no predecessor form when a member departs: pilots who took off together.

pilots	hours	median (min)	join	leave	landing	split	merge
2	17 389	41	0.31 [0.30, 0.32]	0.96 [0.95, 0.97]	0.098 [0.094, 0.10]	–	0.085 [0.079, 0.092]
3	3 349	20	0.54 [0.52, 0.57]	1.6 [1.6, 1.6]	0.050 [0.044, 0.056]	–	0.16 [0.15, 0.18]
4	1 059	13	0.70 [0.66, 0.76]	1.6 [1.6, 1.7]	0.042 [0.032, 0.051]	0.48 [0.45, 0.52]	0.26 [0.22, 0.29]
5	421	11	0.88 [0.80, 0.96]	1.7 [1.6, 1.8]	0.036 [0.021, 0.057]	0.82 [0.76, 0.88]	0.32 [0.26, 0.37]
6	176	8	1.1 [0.93, 1.2]	2.1 [1.9, 2.3]	0.028 [0.011, 0.048]	1.2 [1.0, 1.3]	0.51 [0.40, 0.63]
7	96	7	1.1 [0.91, 1.3]	2.0 [1.8, 2.2]	0.031 [0.0089, 0.074]	1.3 [1.1, 1.5]	0.40 [0.30, 0.53]
8+	107	4	1.8 [1.5, 2.2]	2.5 [2.2, 2.8]	0.028 [0.0084, 0.057]	2.7 [2.4, 3.1]	1.0 [0.73, 1.2]

Table 2: By size: hours spent, median duration of a group, and groups ending per hour by outcome, with 5–95% bands.

Where in the flight. On 1 Hz flights, where the phases are labelled, we take the phase at the instant a flight enters (31 381) or leaves (31 202) a group; entries at the departure (1 179) and exits at the landing (1 358) are counted apart. Both follow the shares of the time alone in each phase (Table 3 in Appendix C): there is no phase where groups preferentially form or break. 34% of the entries are within 10 min of the departure.

When in the day. Group time is concentrated around midday: 52% of a group flight's time at 12 h, 38% at 9 h, 22% at 18 h (Figure 7 in Appendix C).

1.2 What keeps two pilots together

Do two pilots in a group keep the same pace, or are they pulled back to each other? We follow the delay $\ell^*(t)$ inside each contact (Appendix B of the note), leaving out the instants with either pilot in a search. Groups are the 34 082 pair intervals of pilots who took off within 10 km; chance is the 134 202 contacts of the surrogate pairs, two pilots who never saw each other, and the 980 of these that last at least τ^*, selected as the groups are. Three measures:

- $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle$, the mean square change of the delay after s, fitted by an Ornstein–Uhlenbeck process, $D(s) = 2\sigma^2(1 - e^{-ks}) + c$: $1/k$ is how long the delay remembers its value, σ its spread, $2k\sigma^2$ how fast it wanders;
- the drift $\langle\ell(t + 60\text{ s}) - \ell(t)\rangle/60$ s at each ℓ, made symmetric since the order of a pair is arbitrary: the pull back to $\ell = 0$, a spring $-k\ell$ near 0;
- the distance between the two pilots at the same instant after their first contact. Two independent pilots separate as twice the displacement variance of one, since the drift along the route cancels in their difference; the surrogate pairs give this baseline on the same terrain.

- **Groups keep the same pace.** Their delay wanders at $2k\sigma^2 = 21.4$ [20.9, 22.0] s^2/s, against 43 [41, 46] for chance contacts as long and 62 [61, 63] for all of them; its spread is $\sigma = 55$ s [54, 55], against 79 s [78, 80]. Size does not matter: $2k\sigma^2 = 21.2$, 21.4 and 21.8 s^2/s at 2, 3 and 4+ pilots.

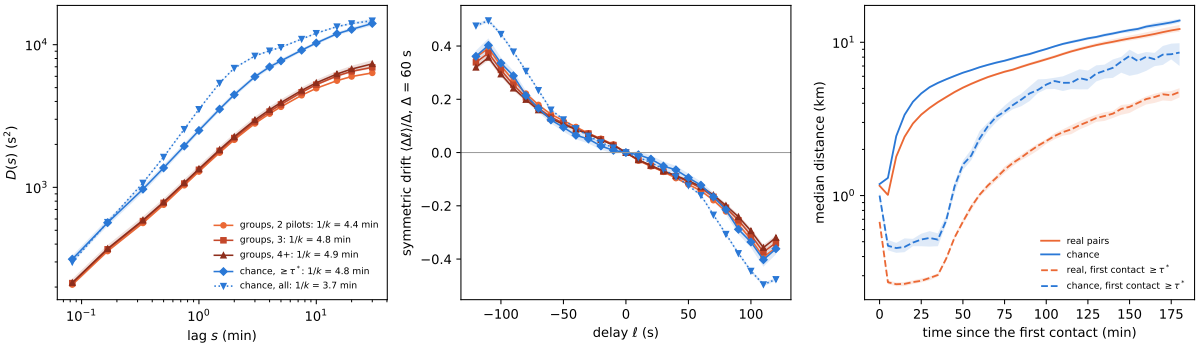


Figure 2: Left: $D(s)$ of groups (by size) and of chance contacts, with $1/k$ of the Ornstein–Uhlenbeck fit. Middle: symmetric drift of the delay. Right: median distance at the same instant after the first contact, for every pair and for those whose first contact lasts at least τ^*. Bands: 5–95% of 200 draws of the dates.

- **They are not pulled back much harder.** The delay forgets itself in $1/k = 4.7$ min [4.6, 4.7], against 4.8 min [4.5, 5.1] by chance; the drift gives 6.9 min [6.7, 7.0] against 7.5 min [6.9, 8.1]. Most of the pull is the selection: a contact that lasts keeps $|\ell| \leq \delta$ by definition.
- **They stay together after the contact that selected them.** When the first contact lasts at least τ^*, the two pilots are 1.0 km apart 60 min later (median), against 2.4 km by chance; 2.9 against 5.8 km after 2 h. Over every first contact, 3.7 against 5.1 km after 30 min.

What holds a group together is a shared pace; a spring adds little to it. Whether the shared pace is following or the same choices in the same air, the data cannot tell.

1.3 Who leads

Who is ahead in a group, how often does that change, and who is it? The answer starts from the shape of the group.

Shape. Inside a group the members are fixed, but the pair intervals among them start and end: each edge set is a configuration. A configuration has a density, a diameter and a shape: complete, a tree (a chain or a star), or between (Appendix A). The TA-MSD pieces of the report *Group and solo flights* lie inside one group and take the shape of its configuration at their middle.

- **Large groups are strings along the route.** Three pilots form a triangle 53% of the time and a chain 47%. Complete graphs fall from 27% of the time at 4 pilots to 2.5% at 8+, the mean density from 0.84 at 3 to 0.45 at 8+; at 8+ the diameter is 4 or more 61% of the time: pilots follow each other along the route.
- **The shape does not change the flight.** $H_{\mathrm{loc}}(1000\ \mathrm{s})$ is 0.906 [0.901, 0.910] in triangles and 0.907 [0.902, 0.911] in chains; 0.915 [0.905, 0.923] in complete graphs of 4+ and 0.907 [0.894, 0.920] in trees. The straighter progress found in *Group and solo flights* does not need every pilot in contact with every other.

Order. The signed median delay ℓ_{med} of a pair interval (Appendix B of the note) says who passes first: the pilot ahead, when the other passes at least 15 s later; otherwise the two fly abreast (33% of the pairs of two; median delay 20 s). How leading changes the flight is in *Group and solo flights*; here we ask who leads (Figure 4; Figure 8 in Appendix C).

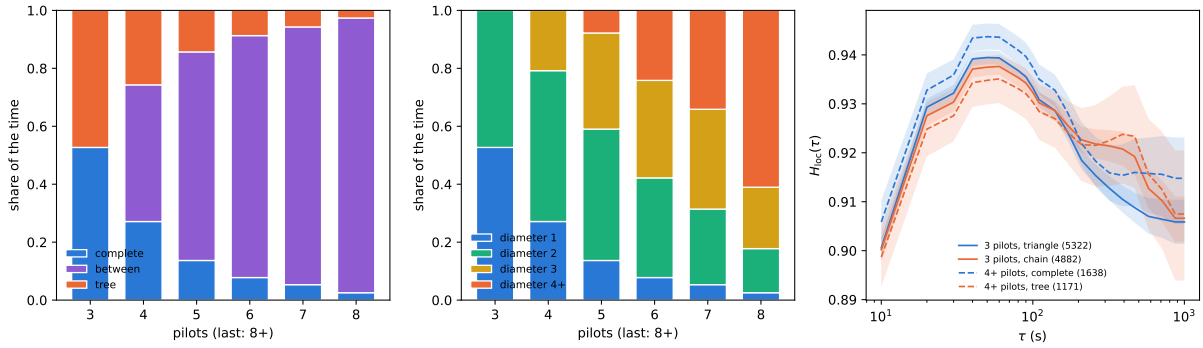


Figure 3: Left: share of the time of each shape, by size (8+ pooled). Middle: share of the time by diameter. Right: $H_{\mathrm{loc}}(\tau)$ of paragliders in triangles and chains of 3 pilots, and in complete and tree graphs of 4 or more (flights in brackets), with 5–95% bands.

- **Pilots are in a line.** In groups of 3 pilots or more, each edge points from the pilot ahead to the one behind; we compare with the same graphs with each direction drawn at random (50 draws). Directed triangles are cyclic 2.3% [1.8, 2.8] of the time, against 25% at random; graphs with every edge directed (34% of the time) are acyclic 98.7%, against 88%. Positions along one route cannot form cycles, so the order is geometry and says nothing about rank.

- **One leader, in small groups:** 80.9% [80.0, 82.0] of the acyclic time, against 77.2% [76.7, 77.8] at random; 86% at 3 pilots, 68% at 4, 52% at 5 or more. A single file (depth $n-1$): 71, 47 and 26%.

- **The lead changes often.** Along $\ell^*(t)$, with stretches shorter than 60 s dropped, the pilot ahead changes 3.3 times per hour of contact [3.3, 3.4], against 4.3 [4.1, 4.4] in chance contacts lasting at least τ^*; the pilot most often ahead holds 74% of the directed time (median), against 69%.

- **The experienced pilot on the faster wing leads.** In groups of two named pilots, the leader has more earlier flights in the catalog in 58.5% [57.8, 59.1] of 16 294 pairs (1.3 times as many, median), and the higher wing class (A = 1 to competition = 5) in 64.3% [63.4, 65.3] of 8 475.

- **Some pilots lead or follow persistently.** For the 975 pilots with at least 10 roles in pairs, the dispersion of their lead fractions is 2.73 against 1.00 with roles swapped at random (95% below 1.08); 12% lead in at most 20% or at least 80% of their roles, against 3% expected.

1.4 With whom

Do pilots choose their partners, or fly with whoever is there? We follow paraglider pilots by their name in the catalog (anonymised ones left out). Two pilots are partners on a day when their flights share a pair interval: a *pair-day*. Chance keeps who flies in a group where and when: within each site-day (takeoffs within 3 km, joined when a pair interval links two sites), the partners are redrawn at random, each pilot keeping its number of partners that day (a configuration model; 100 draws). There are 30 550 pair-days among 4 525 pilots and 22 513 pairs; one pair flew together on 69 days (Figure 5).

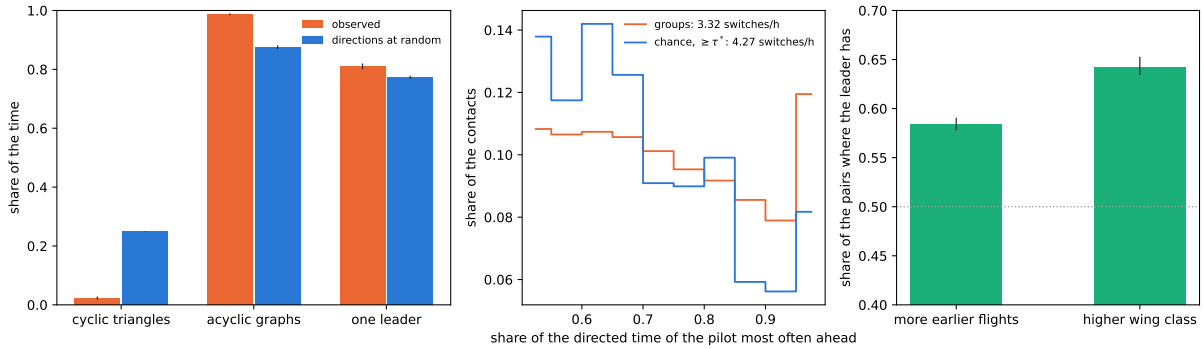


Figure 4: Left: in groups of 3 or more, share of the time with cyclic directed triangles, acyclic graphs (every edge directed) and one leader (acyclic graphs), observed and with directions at random. Middle: share of the directed time of the pilot most often ahead in each contact, with the switches per hour. Right: share of the pairs of two in which the leader has more earlier flights, or the higher wing class. Bars: 5–95% (blocks).

- **Pilots return to the same partners:** 37.6% of the pair-days are on pairs that meet on more than one day, against 25.7% [25.3, 26.0] at random (3 437 such pairs against 2 501).
- **and to their club:** 32.1% of the pair-days join two pilots of the same club, against 24.0% [23.8, 24.2].

Chance already fixes who flies in a group at each site and day, so the excess is the choice of partner among the pilots there. Choosing where and when to fly with friends stays in chance here.

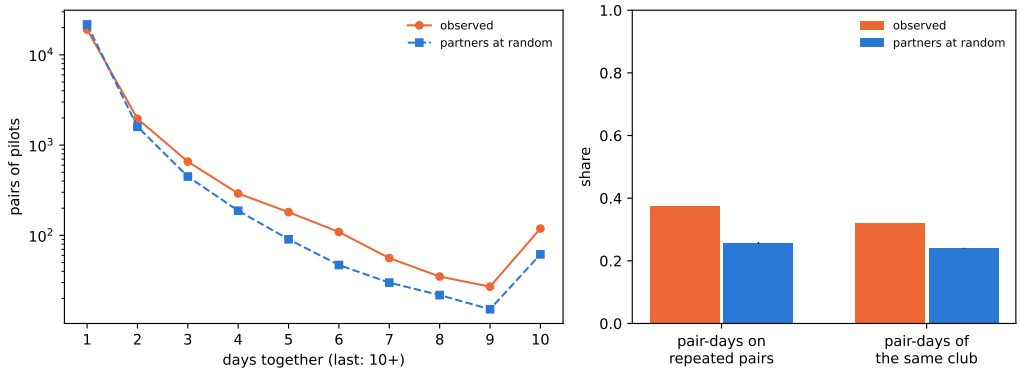


Figure 5: Left: pairs of pilots by the number of days they flew together, observed and with partners at random. Right: share of the pair-days on pairs that meet on more than one day, and of two pilots of the same club (each pilot's main club); bars at random: 5–95% of the draws.

1.5 Where the sizes come from

Why are most groups pairs, and how do a few grow large? $T(n)$ is the share of the time spent in groups of n pilots: the chance that a group seen at a random instant has n pilots. If sizes change one pilot at a time, the flux from n to $n+1$ pilots balances the flux back, a birth–death chain (Appendix B):

$$\frac{T(n+1)}{T(n)} = \frac{g(n)}{l(n+1)}, \tag{1}$$

with $g(n)$ the joins and $l(n)$ the leaves and landings per hour at size n (Table 2); splits and merges are left out. Per member, $g(n) = a\,n$ and $l(n) = \mu\,n^{1-\gamma}$: every member brings in others

at rate a and leaves at rate $\mu n^{-\gamma}$. We fit these to the rates, without looking at T. With $\gamma = 0$ the chain gives the log-series $T(n) \propto x^n/n$, $x = a/\mu$. For comparison, we fit the form of Bonabeau et al. [1], $T(n) \propto n^{-\alpha} e^{-n/n_c}$, directly to T.

- **One pilot at a time is enough.** With the measured rates the chain gives $T(n)$ within 4% from 2 to 8 pilots (0.145 against 0.148 at 3 pilots, just outside its band), although splits and merges are 6% of the changes at 2 pilots and 46% at 8+.
- **Members join at a constant rate and leave less often in large groups.** Each member brings in others at $a = 0.162$ [0.158, 0.167] per hour, at every size. Each leaves at 0.53 per hour in a pair and 0.25 at 8+: $\gamma = 0.30$ [0.28, 0.33].
- **The heavy tail comes from staying.** With $\gamma = 0$ ($x = 0.25$) $T(10)$ is 290 times too small; with $\gamma = 0.30$, 4 times. Fitted directly, $\alpha = 3.4$ and $n_c = 5$.

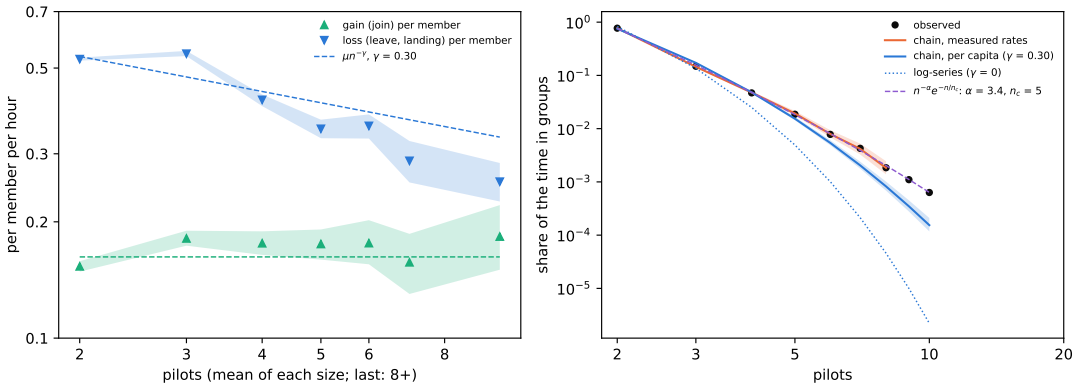


Figure 6: Left: joins and losses per member per hour by size (mean size of each class; 8+ pooled), with a and $\mu n^{-\gamma}$ dashed. Right: $T(n)$ observed (sizes with at least 10 h) and from the chain (1) with the measured rates, with the per-capita rates and with $\gamma = 0$; dashed purple: the form of Bonabeau et al. fitted to T. Bands: 5–95%.

1.6 What the data support

Supported

- groups are mostly pairs; large ones are strings along the route that change members every few minutes, while the episode lasts about an hour;
- group pilots keep the same pace: their delay wanders half as fast as that of two pilots kept in contact by chance, and they stay closer after the contact;
- the size distribution follows from one pilot joining or leaving at a time; its heavy tail from members of larger groups leaving less often;
- the lead changes a few times per hour and goes more often to the experienced pilot on the higher wing;
- pilots return to the same partners and to their club mates more than who flies where and when explains.

Not supported

- a spring that pulls pilots back: the pull is about that of chance contacts selected the same way;
- that the shape of a group (triangle or chain, complete or tree) changes how its pilots fly;
- a social rank from the order in a group: a line along the route is geometry;
- that keeping pace means following: the surrogate pairs fly at another hour, so pilots in the same air at the same time may share the pace without seeing each other;
- pilots are names of the catalog: homonyms join, a pilot who changes name splits.

One caveat: in a thermal the altitude fixes $\ell^*(t)$ to about a turn, 20 s (Appendix B of the note), for real and chance contacts alike, so the cohesion and the switches of the lead carry this noise on both sides of the comparison with chance.

A Graph terms

The terms of graph theory that the sections above use, each with the place where it is used. Graph, path, connected component and directed graph are defined in Appendix F of the note.

Shape (Section 1.3). For a connected graph of n nodes and m edges:

- **Density** $m/\binom{n}{2}$: the share of the node pairs joined by an edge.
- **Distance** between two nodes: the fewest edges of a path joining them. **Diameter**: the largest distance. In the chain 1–2–3 the diameter is 2; in a triangle, 1.
- **Complete graph**: every pair joined, density 1, diameter 1. **Tree**: $m = n - 1$, the fewest edges that keep n nodes connected, with no closed path; a *chain* is a tree with diameter $n - 1$, a *star* one with diameter 2.

Order (Section 1.3). In a directed graph:

- a **cycle** is a path that follows the arrows back to its start (A→B→C→A); a graph with none is **acyclic**, and its nodes can be put in a line with every arrow pointing forward;
- a **source** has no arrow coming in: here, a pilot with nobody ahead, a leader;

- the **depth** of an acyclic graph is its longest chain of arrows: 1 for a star, $n-1$ for a line.

A directed triangle is *transitive* (A→B, B→C, A→C) or *cyclic*; with arrows drawn at random, 2 of its 8 orientations are cyclic.

Configuration model (Section 1.4). A random graph with the *degrees* (the number of edges of each node) of a given one: each node gets as many half-edges as its degree, and the half-edges are paired at random [2]. It keeps how many partners each pilot had and changes only who they were.

B Two models

Section 1 reads two measured quantities through two textbook models.

Ornstein–Uhlenbeck process [3] (Section 1.2). A quantity ℓ pulled back to 0 by a spring of rate k and shaken by noise:

$$d\ell = -k\,\ell\,dt + \sqrt{2k}\,\sigma\,dW. \tag{2}$$

It forgets its value after $1/k$ and spreads by σ around 0. Its mean square change after a lag s is $D(s) = \langle(\ell(t+s) - \ell(t))^2\rangle = 2\sigma^2(1 - e^{-ks})$: it grows as $2k\sigma^2 s$ at short lags, like a free random walk, and stops at $2\sigma^2$ after $1/k$. With $k = 0$ it is a random walk and D grows forever. A measurement noise of variance $c/2$ on each value adds c to $D(s)$ at every lag.

Birth–death chain (Section 1.5). A group of n pilots gains one at rate $g(n)$ and loses one at rate $l(n)$. Over a long time the transitions from n to $n+1$ and back are equally many, so the times spent at the two sizes satisfy $T(n)\,g(n) = T(n+1)\,l(n+1)$, Eq. (1), and multiplying the ratios from 2 to n gives $T(n)$. If every member brings in others at rate a and leaves at rate μ, then $g(n) = an$, $l(n) = \mu n$ and $T(n) \propto x^n/n$ with $x = a/\mu$, the log-series, whose tail falls exponentially. A leaving rate that falls with n makes the tail heavier.

C Phase, time of day and habitual leaders

Table 3, Figure 7 and Figure 8 support Section 1.

	climb (%)	search (%)	transition (%)
entering a group	28.1 [27.6, 28.5]	10.3 [10.0, 10.6]	61.5 [61.0, 61.9]
leaving a group	28.0 [27.6, 28.3]	11.2 [10.9, 11.5]	60.6 [60.1, 61.0]
share of time, alone	29.1 [28.9, 29.3]	10.1 [10.0, 10.2]	60.8 [60.6, 61.0]
share of time, in a group	33.0 [32.8, 33.2]	9.4 [9.2, 9.6]	57.6 [57.3, 57.9]

Table 3: Phase at the instant of entering or leaving a group, and shares of time in each phase (percent, with 5–95% bands; 1 Hz flights).

References

[1] E. Bonabeau, L. Dagorn, P. Fréon. Scaling in animal group-size distributions. *Proceedings of the National Academy of Sciences* 96:4472–4477, 1999.

[2] M. E. J. Newman. *Networks: An Introduction*. Oxford University Press, 2010.

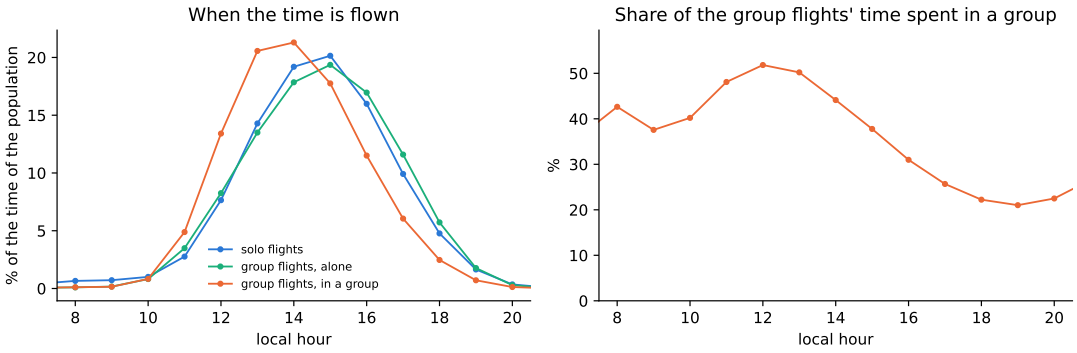


Figure 7: Time of solo flights (S), of group flights alone (A) and in a group (G) over the local hour (left), and the share of a group flight's time spent in a group (right).

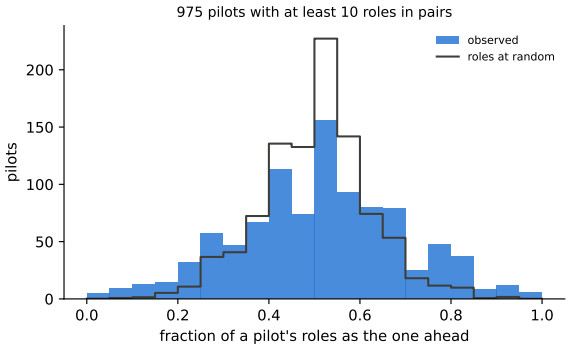


Figure 8: Lead fraction of each pilot, against roles at random.

[3] G. E. Uhlenbeck, L. S. Ornstein. On the theory of the Brownian motion. *Physical Review* 36:823–841, 1930.